COURSE

# Cambridge IGCSE Business (0450 / 0264)

# Stock

**4 Operations management · 4.1 Production of goods and services: inventory and lean production**

Revision for 4 Operations management: each firm works it out on its own deals.

Class **Cambridge IGCSE Business (0450 / 0264)** Topic **Stock**

## THE LESSON

| | |
|---|---|
| **Setup** | 3 min |

**Round 1** — Decisions 3 min · Deals 5 min · Topic tasks 4 min

| | | | |
|---|---|---|---|
| **Firm decisions** | **Deals** | TASK 1 **Stock holding cost** | TASK 2 **Too much stock** |

**Round 2** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| | | | | |
|---|---|---|---|---|
| **Firm decisions** | **Deals** | TASK 3 **Rank supplier cost** | TASK 4 **Saving from JIT** | **B2B deal** |

**Round 3** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| | | | | |
|---|---|---|---|---|
| **Firm decisions** | **Deals** | TASK 5 **JIT or hold stock** | TASK 6 **JIT vs discount** | **B2B deal** |

| | |
|---|---|
| **Final questions** | 3 min |

Fits a 30-minute lesson setting; longer settings add more deals.

# Each step

| | |
|---|---|
| **Firm decisions** | Each role makes its decision on its own page. |
| **Deals** | Each pupil trades with a classmate in another firm. |
| **Topic task** | The firm answers together. Everyone presses Agree. |
| **B2B deal** | One firm sells a lot of cars to another firm. |
| **Final questions** | Each pupil answers six questions. |

# Firm decisions

| | | |
|---|---|---|
| **CEO** | **Staff** | Hire staff. Each costs a salary and gives one benefit. |
| | **Sabotage** FROM ROUND 2 | Hit a rival firm's cash, at a cost. A Legal Counsel blocks it. |
| **Sales & Marketing** | **Marketing** | Book one advert campaign to lift the next sales. |
| **Finance** | **Capital & Loans** | Borrow cash. The firm pays it back with interest. |
| | **Investments** FROM ROUND 2 | Buy property or funds. The value can rise or fall. |
| **Operations** | **Seasonal stock** | Order the season's cars in advance. |

**B2B deal · Rounds 2 and 3**

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

# Topic tasks

Each firm gets its own figures. The method is the same for every firm.

## Task 1

ROUND 1

Cars wait in your yard.

**Cost of holding the cars?**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| You hold for them | **8 cars** |
| Holding cost per car a week | **£100** |
| Weeks you hold them | **4 weeks** |

METHOD

**Holding cost = cars × cost per car a week × weeks**

8 × £100 × 4 = £3,200

**Why:** Stock that waits costs £800 every week.

## Task 2

ROUND 1

Your yard holds far more cars than you sell in a month.

**Which two problems are most likely?**

PICK TWO

- ✔ **Cash tied up in stock**
- ✔ **Higher storage costs**
- ○ Lost sales from empty space
- ○ Lower insurance costs
- ○ Running out of cars to sell

**Why:** Extra stock holds cash and costs money to keep.

## Task 3

ROUND 2

Three suppliers quote for 10 cars, delivered.

**Order by total cost, cheapest first?**

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **£8,100 a car, £1,200 delivery**
2. **£8,000 a car, £2,500 delivery**
3. **£8,300 a car, £300 delivery**

**Why:** £82,200, £82,500 and £83,300: the lowest price per car is not the cheapest order.

## Task 4

ROUND 2

Just in time, or in case?

**Saving a month from JIT?**

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

| | |
|---|---|
| Spare cars held, just in case | **7 cars** |
| Holding cost per car a week | **£70** |
| Extra JIT deliveries, a week | **£200** |
| Weeks in the month | **4 weeks** |

METHOD

**Saving = weeks × (spare cars × holding cost − extra deliveries)**

4 × (7 × £70 − £200) = £1,160

**Why:** JIT saves the holding cost, but the extra deliveries cost £800 a month.

## Task 5

ROUND 3

Your supplier is two hours away and always on time. Hold 40 cars, or hold 10 and reorder often?

**Which should you choose, and why?**

PICK A OR B AND GIVE A REASON

- ✔ **Hold 10 and reorder often**
- ○ Hold 40 cars

**Why:** A reliable, nearby supplier lets you hold less and free up cash.

## Task 6

ROUND 3

JIT loses a bulk discount.

**Gain a month from JIT? Minus allowed.**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| Cars bought a month | **13 cars** |
| Price per car | **£8,000** |
| Bulk discount lost with JIT | **3%** |
| Holding cost saved a month | **£2,000** |

METHOD

**Gain = holding cost saved – cars × price × discount lost**

13 × £8,000 × 3% = £3,120; £2,000 − £3,120 = −£1,120

**Why:** Smaller, frequent orders save holding costs but can lose bulk-buying discounts.

# Final questions

**6**
questions for each pupil

**159**
Level 2 Business questions

**4**
of them on stock

**The game draws 6 of the 159 Level 2 questions; these 4 are about stock.**

1 What is an opportunity cost of holding too much vehicle stock?
**The cash cannot be used for anything else.**

2 A workshop keeps almost no spare parts and orders only when needed. What is the main risk of this just-in-time approach?
**A late delivery can stop work.**

3 High-value parts repeatedly disappear from stock records. Which control system is most useful?
**Restrict access, log movements and reconcile counts with records.**

4 A supply chain is every business that helps get a product to the customer. A tyre maker's factory floods. Using this, what happens to a workshop?
**Its tyre deliveries may stop, though it did nothing wrong.**